BIOT 6900 · Module 3 Assignment

Target Triage with Protein Language Models: Alzheimer's Disease

Sharath Mandula | Computational Lab 3 | Fall 2026 | Disease: Alzheimer's disease (AD)

Overview

In Module 2, I ranked Alzheimer's disease genes using three layers of evidence: RNA (GSE5281 hippocampus), protein (Johnson et al. 2022 TMT proteomics) and genetics (GWAS Catalog). After correcting the scoring so that each layer is rank-normalised before equal weighting, 1,139 genes survived the three-way join. In this module, I took the top 15 of those genes and added one more line of evidence: **does each target look like proteins we already know how to drug, and by which modality?**

For this, I used the pre-computed ESM C (300M) embeddings from the data pack, which convert every protein sequence into 960 numbers. A logistic regression trained on a panel of 1,201 human proteins with Open Targets clinical precedence labels (approved, advanced clinical or Phase 1) turns these embeddings into a tractability prior. I also looked at zero-shot variant scores from ESM-2 (650M) for the course anchor variants.

Mapping and leakage. All 15 of my targets mapped to a reviewed protein in proteome.tsv, so there were **no unmapped genes** to report. All of them are protein-coding; if a non-coding RNA had come through, it would have needed a different kind of evidence, such as eQTL or regulatory data, because there is no protein sequence to embed. For the leakage guard, one of my targets, **CACNG8**, was already in the training panel with a small-molecule clinical label. I removed it before fitting, so the model trained on 1,200 proteins and no target was predicted using its own label.

1. Re-ranking: did the prior agree with my Module 2 ranking?

Mostly, no. The Spearman correlation between my Module 2 score and the best-modality tractability score across the 15 targets was only about 0.14. This is not surprising, because the two scores answer completely different questions. Module 2 asks "is this gene changed or associated in AD?", while the prior asks "does this protein look like something that has been drugged before?" Table 1 shows the final ranking.

Final rank	Gene	Module 2 rank	Module 2 score	p_sm	p_ab	Final score
1	SAMHD1	1	0.933	0.824	0.003	1.000
2	CACNG8	6	0.867	0.380	0.001	0.773
3	ASAP1	5	0.871	0.193	0.002	0.760
4	ZMYM2	3	0.896	0.040	0.004	0.680
5	PDZD2	2	0.898	0.020	0.002	0.667
6	APOC1	9	0.859	0.001	0.248	0.627
7	COL11A1	7	0.863	0.100	0.000	0.600
8	APOE	4	0.891	0.002	0.016	0.560
9	NDUFA10	11	0.856	0.055	0.003	0.387
10	CLU	13	0.841	0.036	0.131	0.387
11	WNK2	12	0.843	0.057	0.017	0.373
12	FARP2	14	0.840	0.180	0.001	0.373
13	RHOJ	10	0.856	0.034	0.001	0.373
14	MAPT	8	0.860	0.006	0.001	0.347
15	NUMA1	15	0.835	0.008	0.005	0.093

Table 1. Final re-ranked targets (w = 0.6). p_sm and p_ab are the small-molecule and antibody priors.

The top stayed the same: **SAMHD1** was first in Module 2 and also has the highest small-molecule prior (0.824), so both lines of evidence agree on it. The biggest mover was **MAPT (tau), which dropped from 8th to 14th**. Both of its priors are close to zero (p_sm 0.006, p_ab 0.001). Tau is an intrinsically disordered protein with no stable pocket, and its nearest drugged neighbours are only moderately similar (PDE4D, HCN4, CACNB2 at about 0.87–0.88), so it does not resemble any classic drug target family. Interestingly, anti-tau antibodies such as semorinemab and gosuranemab have been tested in clinical trials, so the very low p_ab here shows that the prior only knows what is common in its training labels. APOE also fell from 4th to 8th for a similar reason. On the other side, **CACNG8 rose the most, from 6th to 2nd**. Even with its own label hidden, it scored 0.380 for small molecules because its nearest drugged neighbours are ion channel proteins (HCN4, KCNQ4, PDE4D). This fits its biology: CACNG8 (TARP γ-8) is an auxiliary subunit of hippocampal AMPA receptors, and selective modulators of γ-8 containing receptors have been developed as anti-seizure drugs.

Choice of w = 0.6. I gave 60% weight to the Module 2 evidence and 40% to tractability. My reasoning is that Module 2 is specific to Alzheimer's disease and combines three independent data types, whereas the prior is generic and only moderately accurate on new families (family-split AUROC 0.728, see Section 3). So disease evidence should lead. At the same time, I did not want to make w too high, because a disease gene with no realistic way to drug it is less useful for a discovery team. One point I noticed is that the Module 2 scores of my top 15 are very close to each other (0.835 to 0.933), so after converting them to percentile ranks, even 40% weight on tractability is enough to reshuffle the middle of the list. If I were less confident about the Module 2 data, which comes from different cohorts and brain regions, I would argue for w = 0.5.

2. Modality: where p_sm and p_ab differ sharply

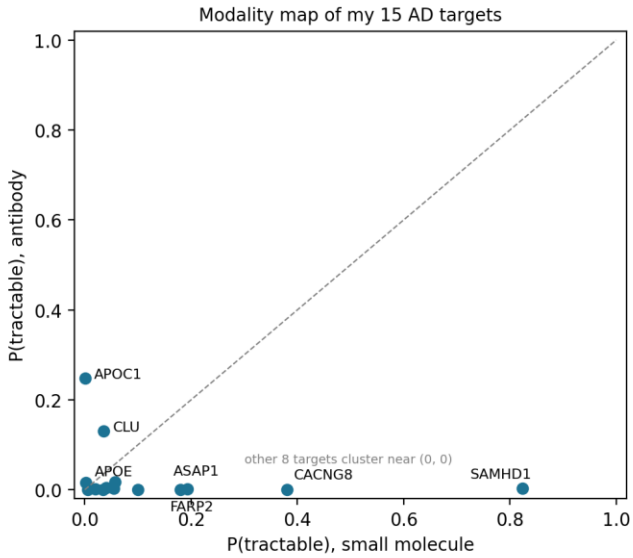


Figure 1. Small-molecule versus antibody prior for my 15 targets. The dashed line marks equal priors.

The clearest contrast in my list is between **SAMHD1** and **APOC1**. SAMHD1 has p_sm = 0.824 but p_ab = 0.003. SAMHD1 is an intracellular enzyme (a dNTP triphosphohydrolase) that controls the cell's pool of DNA building blocks; it is known for restricting HIV-1 in myeloid cells, and mutations in it cause Aicardi–Goutières syndrome. Because it sits inside the cell and works by binding nucleotides, it has well-defined catalytic and allosteric nucleotide-binding sites, which is exactly the kind of pocket small molecules can occupy. Its nearest drugged neighbours (PDE1B, CPT1B and

PARP1, all around 0.95 similarity) are also enzymes with small-molecule drugs. An antibody, on the other hand, cannot reach an intracellular enzyme, so the low p_ab makes biological sense.

APOC1 shows the reverse pattern: p_ab = 0.248 and p_sm = 0.001. APOC1 is a small (83 amino acid), secreted apolipoprotein that circulates on lipoprotein particles. It has no enzyme active site, but because it is extracellular, it is reachable by an antibody. CLU (clusterin) shows the same direction (p_ab 0.131 versus p_sm 0.036) and is also a secreted chaperone. **APOE** fits this biology as well, being a secreted apolipoprotein, but both of its priors are very low (0.016 and 0.002). I think this is because very few apolipoproteins have clinical-stage drugs, so the model has nothing similar to learn from. This is a good reminder that a low prior means "nothing like this has been drugged before", not "this cannot be drugged".

3. Validation: family-split metrics

On the training panel, the random split gave AUROC 0.832 and average precision (AP) 0.706, while the **family split gave AUROC 0.728 and AP 0.531**. The chance level for AP is 0.333, because about one third of the panel has clinical precedence. So the prior is clearly better than chance even on held-out families, but it is weaker than the random split suggests.

The family-split number is the honest one. In a random split, close relatives from the same protein family (for example, two kinases or two GPCRs) end up on both sides, one in training and one in testing. The model then only needs to recognise the family, not learn anything general, and the score gets inflated. This is homology leakage. In real target discovery, the interesting case is a new protein from a family that is not already full of drugs, which is exactly what the family split simulates by holding out whole families. The random-split number would have hidden a drop of about 0.10 in AUROC and about 0.18 in AP, and it would have made my tractability scores look more trustworthy than they actually are. This is also part of why I kept w above 0.5.

4. Interpreting the anchor variant scores

Gene	Variant (UniProt numbering)	Note	Score	Percentile in protein
APOE	R176C (mature R158C)	rs7412, defines ε2	−8.56	11.8
APOE	C130R (mature C112R)	rs429358, defines ε4	+7.21	100.0
TREM2	R47H	rs75932628, AD risk	−5.16	47.7
TP53	R175H	cancer hotspot	−5.97	19.3

Table 2. Zero-shot variant scores (log P(mutant) − log P(wild-type)). I use UniProt precursor numbering; APOE mature numbering is 18 lower.

These results show clearly that a negative score is not the same as pathogenic. The **APOE ε2 variant (R176C) scored −8.56**, in the 11.8th percentile of all possible changes in APOE, so the model finds it quite surprising. But ε2 is actually the protective allele for Alzheimer's disease. The opposite case is even more striking: **APOE ε4 (C130R), the strongest common genetic risk factor for late-onset AD, scored +7.21 and sits at the 100th percentile**, meaning the model considers it the most acceptable substitution in the whole protein. The reason is evolutionary. Arginine at this position is the ancestral state, shared with other primates, so a model trained on natural sequences sees it as completely normal. It cannot know that this "normal" residue increases AD risk in modern humans.

TREM2 R47H scored −5.16, right in the middle of the distribution (47.7th percentile), even though it raises AD risk roughly two to three fold. For comparison, the **p53 hotspot R175H** scored −5.97 (19.3rd percentile). It is in the damaging tail, but not at the very extreme, even though it is one of the most common cancer drivers. R175H disrupts zinc binding and misfolds the DNA-binding domain,

and cancer cells are selected for this change precisely because it is harmful to normal p53 function, which is different from what evolution in healthy organisms has filtered out.

The main point is that these scores measure **evolutionary plausibility**, not clinical effect. The model only sees a single missense change in one sequence. It cannot see the effect in a specific tissue or cell type, gain-of-function or dominant-negative effects, interactions with other proteins or lipids, the combination of two SNPs that defines an allele (ε2, ε3 and ε4 are defined by two positions together), population frequency, or environmental factors. It also cannot score anything that is not a missense change, such as splicing, regulatory, or non-coding variants.

5. Limitations and next step

Sequence alone cannot tell me whether a target actually a druggable pocket has, how big it is, or whether it is open in the protein's real conformation. It also does not directly tell me localisation, expression in the right cell type (for example, microglia versus neurons), or whether the target is part of a complex. The labels themselves encode history: they reflect which proteins pharma has chosen to work on so far, not which proteins are truly druggable, and the training panel is enriched (about one third positives), so the probabilities should be read as rankings, not as real chances. There is also a technical issue in my list: **PDZD2, WNK2 and NUMA1 are longer than the 2,046-residue limit**, so their embeddings were built from truncated sequences. This may explain why WNK2 has a low small-molecule prior (0.057) even though it is a protein kinase; its kinase domain is only a small part of a very long protein.

For Week 4 structure-based pocket assessment, I would send these two targets:

SAMHD1 is my top-ranked target after re-ranking, with both the strongest Module 2 evidence and the highest small-molecule prior. It is an intracellular enzyme with experimental structures available, so Week 4 can check whether AlphaFold2 and ESMFold reproduce its nucleotide-binding pockets with good confidence, and whether those pockets score as druggable. It would also be important to look at its link with AD more carefully, since its high Module 2 score came mainly from strong RNA down-regulation.

APOE is the most important genetic target in Alzheimer's disease, but the sequence prior scored it near zero. Structure is the best way to test whether this is a blind spot of the prior. In particular, the ε4 form has a domain interaction between the N- and C-terminal regions, and small molecules that block this interaction ("structure correctors") have been reported in preclinical studies. A structure-based pocket search could show whether APOE has druggable sites that a sequence-only model simply cannot see.

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